

Vision-Based Navigation for AMRs in a Dynamic Warehouse

Progress summary for committee review

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1. The question

Can a vision-based pipeline provide reliable localization for an autonomous mobile robot in a warehouse that contains moving objects? The work compares two stereo-inertial front ends, ORB-SLAM3 and VINS-Fusion, under controlled changes in scene dynamics, floor texture and compute load, and asks whether masking dynamic objects with a learned detector improves localization. Path planning is outside the scope: the deliverable is a reliable navigation state and map, not a planner.

2. What exists

A warehouse simulator with static, dynamic, textured-floor and featureless-floor variants; both estimators instrumented to record per-frame timing, delivery, state, CPU and memory; a custom YOLO detector feeding dynamic-object masks into both; RTAB-Map as a separate mapping and loop-closure back end; and a run-logging and validity-audit harness. The evidence base is 66 logged estimator runs across seven simulation datasets at three repeats per condition, five multi-sensor recordings spanning four worlds, 9860 camera frames with projected object ground truth, and one reconstructed map with geometry and drivability evaluation.

3. What is established

In static scenes the comparison is decided by the noise model, not the architecture. A wheel+IMU EKF achieves the lowest aligned ATE on four of five recordings. That ranking does not survive a control: the EKF's noise model is measured from each recording, while both visual pipelines read four constants from a hand-set file. Given that file, the filter loses to VINS-Fusion on *all five*.

Dataset	IMU only	Wheel odom.	VINS-Fusion	ORB-SLAM3	EKF	EKF [†]
allsensor_000	21.70	2.80	0.62	0.58	0.19	1.31
allsensor_001	12.62	0.98	0.38	0.37	0.31	0.68
allsensor_002	13.24	1.49	0.56	0.60	0.26	0.87
allsensor_003	45.56	3.19	0.78	6.48	0.42	1.23
allsensor_004	26.71	3.24	0.43	1.24	0.49	1.58

Aligned ATE RMSE in metres, rigid $SE(3)$ fit, scale fixed. †: the same filter given the visual pipelines' four IMU noise terms.

The project now has the proprioceptive baseline it previously lacked, and it is a floor the visual system must clear. But the height of that floor is a tuning result: neither visual pipeline has ever been tuned to this platform, the cheapest untried improvement available. The advantage is also specific to these conditions — static scene, simulated encoders, slip under 1% — so it sharpens the thesis question rather than answering it: vision must justify itself where dead reckoning fails, which means long runs, wheel slip, relocalization after tracking loss, and moving obstacles.

The detector's limit is detection, not semantics. Over 9860 frames the detector reaches precision 0.684 and moving-object recall 0.591, while class assignment is 0.996 accurate once an object has been localized. Improving the classifier would change nothing; recall is the binding constraint. The model also produces bounding boxes rather than segmentation, and class membership is not evidence of motion.

The reconstructed map is geometrically sound. Against the simulator's own collision geometry at 0.25 m tolerance, occupied-cell precision is 96.6% with 89.5% recall over the region the robot observed. The lower whole-building figure of 67.2% is route coverage — the run never entered the side aisles — not mapping failure.

Neither pipeline runs at input rate. Frames arrive every 33 ms; ORB-SLAM3 tracking takes 63 ms and its local mapping 190 ms, and enabling the detector raises these to 113 and 356 ms. Input retention falls from 89% to 67%.

4. What is not established

The dynamic-scene results are withheld. Those worlds are not yet a physical simulation: props follow scripted kinematic paths, carrying collision geometry but producing no dynamic response against the robot. Forty-eight runs were logged there and all forty-eight fail the plausibility check, but those numbers describe the simulation rather than the problem, so they are excluded here and retained in the record as excluded evidence. The consequence is that the condition this thesis is named for has *no valid testbed* — untested, rather than tested and failed. Rebuilding those worlds is the first item of the next period and the cheapest candidate explanation for the breakdown itself.

The core hypothesis has no valid test behind it. Two instrumentation faults prevent it, and the environment question above compounds them. The VINS-Fusion filter never fired: all twelve runs requested it and all twelve record zero matched detector frames and zero removed tracks, even though the detector works offline. Separately, eleven of twelve ORB-SLAM3 baseline/filtered pairs received frame counts differing by 10 % to 30 % on the same recording. One admissible pair exists out of twenty-four. No cross-treatment accuracy conclusion is drawn. Both are code-path faults and stand regardless of scene, but the filtered runs were recorded in the dynamic worlds, so the hypothesis now needs a working mask path *and* a valid environment.

5. How the work is validated

Every comparison is tested for admissibility before its numbers are read: the same recorded input to both treatments within 5 %, the intended treatment demonstrably active rather than merely requested, a common ground-truth interval with the matched sample count reported, and physical plausibility. Five claims were retracted or prevented this period by that protocol — a camera-rotation parsing bug found through map distortion while ATE looked plausible throughout; a map wrongly judged unnavigable because its grid was mis-seated; a drivability score measured against the map’s own construction run; the two filter faults above; and the accuracy ranking in section 3, the project’s most favourable result, overturned by a control run against it deliberately. Each would otherwise have entered the thesis as a result.

6. Next period, in priority order

1. Repair the VINS mask delivery path. This alone unblocks the core hypothesis, and needs no additional data.
2. Equalise ORB-SLAM3 input across paired runs so the filter comparison becomes admissible.
3. Rebuild the dynamic worlds with physically simulated props, giving the titular condition a valid testbed.
4. Add disturbance to the simulation — wheel slip and encoder quantisation — since both idealisations currently favour the dead-reckoning baseline.
5. Record an independent route for map drivability, and extend the map evaluation to the large warehouse.

7. Guidance requested

1. **Scope.** The static-scene comparison reduces to a tuning difference. Should the next period equalise it — tuning both visual pipelines to this sensor suite and re-running — before any decision on whether the thesis argues for vision as the primary localization sensor?
2. **Depth or breadth.** Rebuild the dynamic environment properly before measuring anything further in it, or extend to the real robot and the large warehouse with the static results as they stand?
3. **Integration.** Relocalization, global optimization and multi-session map merging are listed as future work. Is that the right next contribution, or premature until dynamic-scene tracking is sound?

Full detail, including every retracted, excluded and failed measurement, is in the accompanying progress report. The 77 passages that are measured but not yet trustworthy as evidence are listed in the separate Open concerns document, and are excluded from the claims above.